Session plan 7: Predictive inference

Thu 15.10.26, 16:15–17:45 · Room 3.B48 · Basic readings: Kleinberg, Ludwig, Mullainathan & Obermeyer (2015); Hofman et al. (2021), pp. 181–184; additional: Grimmer, Roberts & Stewart (2021) · Applied reading: Ashokkumar, Hewitt, Ghezae & Willer, *Large language models can predict the results of social science experiments* (accepted *Nature* manuscript; syllabus: Hewitt et al. 2024) · Presenters: **Noa Flückiger, Malte Jonas Kohm**

Learning goals: students can (1) tell prediction problems from causal problems (umbrella vs. rain dance); (2) explain why a bad cause can be a good predictor; (3) say when predictions break and why a baseline matters; (4) explain why research should be judged on both explanation and prediction (Hofman et al.); (5) name uses of prediction as a step towards measurement or design.

Before class: take-home exercise I uploaded to OLAT (beginner + advanced PDFs, materials folder; **not** the solutions) · presentation slides in OLAT (due Wed 14.10, 12:00)? · seminar questions 7 uploaded? · grading sheet `session07_handwritten.pdf` ready · deck in speaker view (press S)

Checkpoints: 16:35 start "When predictions break" · 16:49 start pairs exercise · **17:07 hand over** · **17:08 presentation starts** (25 min max., then discussion until 17:42). About 6 min buffer before 17:07

Time	Slide	Content	Interaction	Guiding questions	If behind
16:15–16:17	2	Recap session 6: potential outcomes, counterfactual, four patterns	Input		
16:17–16:22	3	**Umbrella or rain dance?** Four decisions (Kleinberg et al. 2015)	Classify A–D, then seminar question 7.1	Which decisions need a causal answer, which only a prediction?	Skip 7.1 answers
16:22–16:25	4	Predictive inference: unobserved outcomes; success = accuracy on new cases	Input		
16:25–16:30	5	**Hip replacements**: predict futile surgery (Kleinberg et al., pp. 4–5)	Question to the room	Which variables? Must they be causes? What could go wrong?	
16:30–16:35	6	**The four patterns are fine now**: ice cream, umbrellas, age	Seminar question 7.2	Why is a bad cause a good predictor?	
16:35–16:39	7	**When predictions break**; Google Flu and baselines (Hofman p. 182)	Question	Does the turnout model still work if the campaign changes what it does?	
16:39–16:43	8	**Explanation or prediction?** Two traditions; judge research on both (Hofman pp. 181–183)	Question	Can a study explain well but predict badly? The other way round?	2 min
16:43–16:46	9	Prediction as a step: measurement, design (Grimmer et al. 2021)	Question	Where could prediction help in your project?	Skip question
16:46–16:49	10–11	LLMs predict the next word; could they predict experiments? (optional autocomplete demo)	Students guess	How well would an LLM do?	Skip demo
16:49–16:57	12	**Pairs**: own prediction problem, good predictor/bad cause, what could break it	5 min pairs, 3 min share		No sharing
16:57–16:59	13	Three inferential goals	Input		
16:59–17:01	14	**Take-home exercise I** (deadline 29.10.26)	Admin		
17:07–17:08	15	Hand over			

Time	Slide	Content	Interaction	Guiding questions	If behind
17:08–17:33	—	**Student presentation** (25 min max.): summary and critique ~10 min, alternative design ~15 min	Presentation; grade with the sheet		Signal at 20 min
17:33–17:42	—	**Discussion** to improve the proposal (prompts below)	Plenary	See prompts below	
17:42–17:45	16–18	Session 8 on 22.10, 14:15, room 4.B51 (AJ ch. 1; Hüsler & Jäggi); seminar questions 8–9 due Wed 21.10, 12:00	Admin		

Discussion prompts (if the room is quiet): What does the LLM actually "simulate" — participants, or what researchers expected? · Effects are overestimated about twofold: what does that mean for using LLMs in power analysis? · Which hypothesis would you pilot with an LLM, and which never? · Does it travel beyond US survey experiments with text stimuli (the authors' own limitation)? · Does the paper explain anything, or only predict?

Grading (0–5 each, 15/30 = pass): clarity · understanding · critique quality · reasoning · feasibility · discussion. Both presenters, one grade.

Notes during class: